

Henry Rao

Embedded Systems & IoT Firmware Engineer • 4 Years Experience • MERIQ Index: 94% (Mastered)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: FreeRTOS, Python, I2C/SPI/UART, PCB Debugging, MQTT, ARM Cortex, Embedded Linux

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Embedded Systems & IoT Firmware Engineer — *Texas Instruments* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core FreeRTOS service, reducing latency by 28%.
- Led a squad of 5 engineers delivering scalable microservices processing 46M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 38%.

Associate Embedded Systems & IoT Firmware Engineer — *STMicroelectronics* (2022 - 2023)

- Designed and maintained Python data endpoints with 96% SLA uptime.
- Optimized database indexing and caching layers using I2C/SPI/UART saving \$25k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [FreeRTOS, Python, I2C/SPI/UART, PCB Debugging]
Engineered end-to-end production solution leveraging FreeRTOS and Python with full test automation.

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [I2C/SPI/UART, PCB Debugging, MQTT, ARM Cortex]
Architected high-resilience system integrating I2C/SPI/UART with 99.9% uptime.

EDUCATION & CREDENTIALS

B.Tech in Computer Science and Engineering

IIIT Hyderabad — 2018 - 2022 (Score: 8.5 / 10.0)